difference in the approach with respect to k-means is already clear from its name: while k-means is a centroid-based algorithm, and as such builds clusters as convex sets around points elected as centroids, a density-based algorithm defines clusters as sets of points that are close to each other, close enough that the density of points in any area of a cluster is above a certain threshold. The natural extension of this definition, by the way, introduces the concept of noise (also referred to as *outliers*), for those points that are in low-density regions: we will formally define both categories in a few lines, but first, we still have a few high-level considerations on the algorithm.

Like k-means, DBSCAN is a flat hard-clustering algorithm, meaning that each point is assigned to (at most) one cluster (or no cluster, for outliers) with 100% confidence, and that all clusters are objects at the same level, no hierarchy of these groups is kept.

In k-means random initialization of the centroids has a major role in the algorithm (with good choices speeding-up convergence), so much so that often several random restarts of the algorithm are compared before choosing the best clustering; this isn't true for DBSCAN, where points are cycled through somehow randomly, but this has a lower influence, if any, on the final result: therefore, this algorithm can be considered deterministic[12].

DBSCAN, instead, extends the concept of *single-linkage clustering*[13] (*SLC*) by introducing a minimum points-density required to consider two points connected to each other: this reduces the *single-link chain effect*, the worst side-effect of